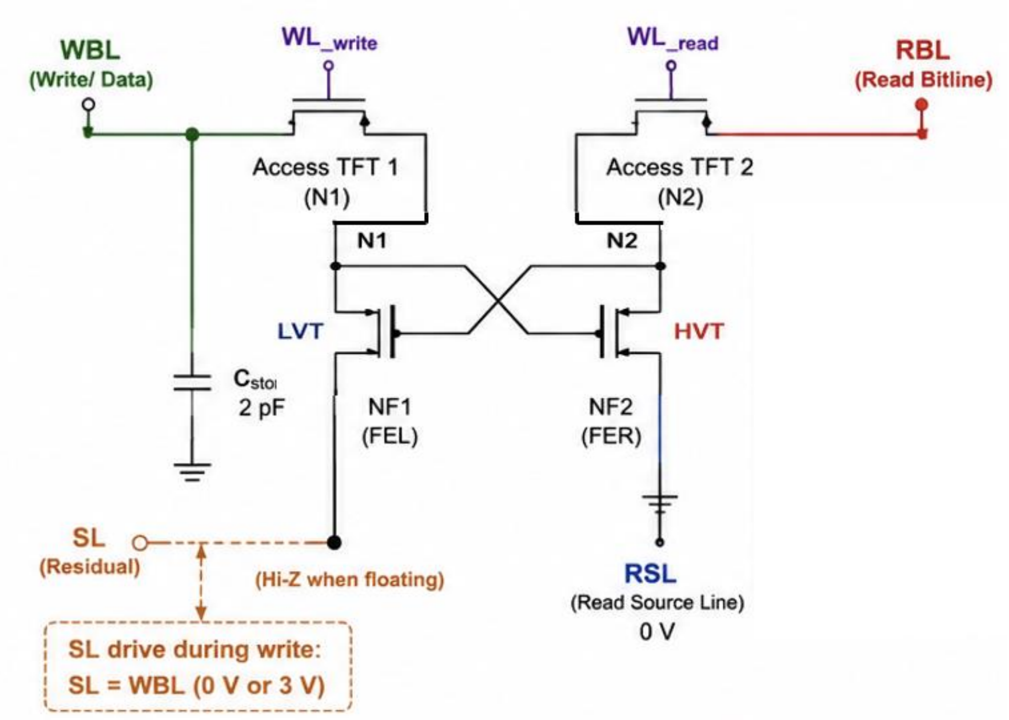
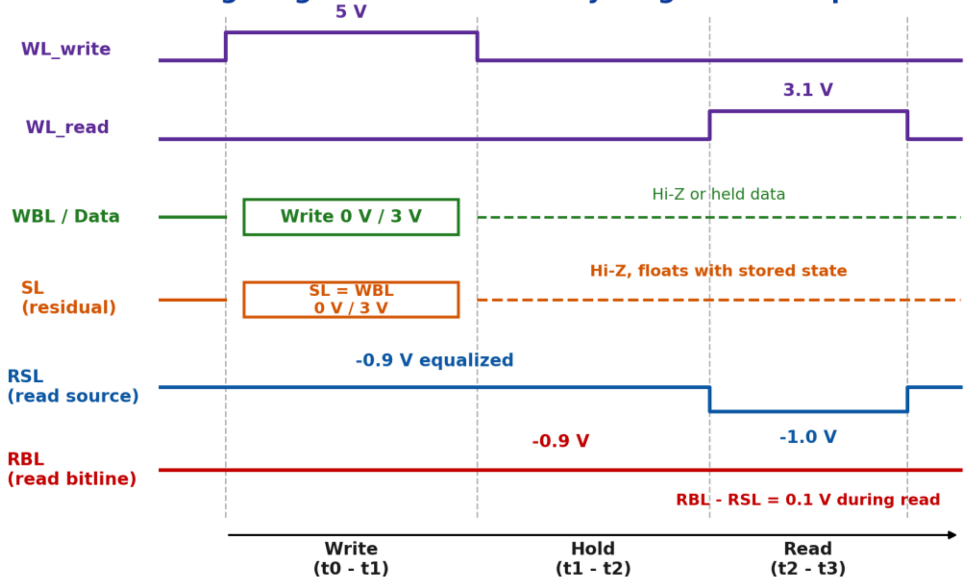
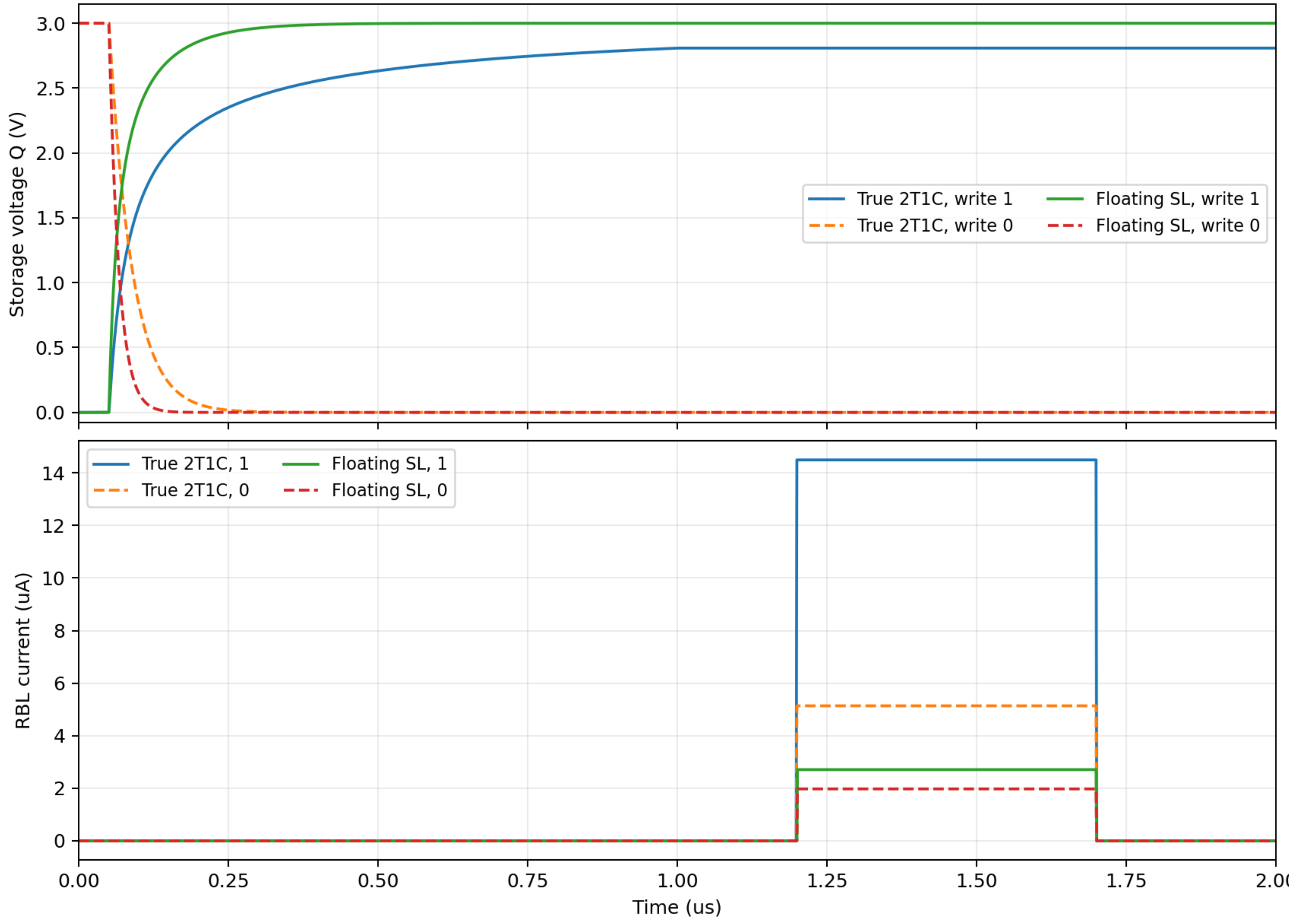
Cross-Coupled 4T Cell (for Single-Ended Read)



Timing Diagram - Best Bias-Only Single-Ended Operation



True 2T1C vs floating SL at best bias-only point



Bias Conditions - Best Bias-Only Point

Phase	WL_write	WL_read	WBL	SL	RSL	RBL
Write	5 V (ON)	0 V (OFF)	0 V or 3 V (data)	= WBL (0 V or 3 V)	-0.9 V	-0.9 V (equalized)
Hold	0 V (OFF)	0 V (OFF)	Hi-Z or held data	Hi-Z (floating)	-0.9 V	-0.9 V (equalized)
Read	0 V (OFF)	3.1 V (ON)	Hi-Z or held data	Hi-Z (floating)	-1.0 V	-0.9 V (read bias)

Read condition: RBL - RSL = 0.1 V. Outside read, RBL and RSL are equalized at -0.9 V so the read path has zero VDS.

Wider N2

At original read bias, `RBL=0.1 V`, `RSL=0 V`, `WLread=4 V`
So with no read-bias change, best simulated point is **`N2=64x (N1=64x for symmetry purpose only)`** , margin **3.518 uA**.